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Faculty of Exact Sciences and Engineering

Data Science Project

Mid-journey Report

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I. INTRODUCTION

Urban transportation systems generate vast amounts of data that hold critical insights for optimizing services, improving pricing strategies, and enhancing operational efficiency. The New York City Taxi Trips dataset, comprising over 7.6 million records from 2019, serves as a rich resource for analyzing taxi trip patterns, fare determinants, and demand dynamics. Each trip record includes features such as pickup/drop-off timestamps, passenger counts, trip distances, fare components, and geographic identifiers, offering a granular view of taxi operations in one of the world's busiest cities. Accurate fare prediction and classification are pivotal for stakeholders—passengers benefit from transparent pricing, while service providers gain tools for fraud detection, demand forecasting, and resource allocation.

This project addresses two interconnected tasks:

- **Regression:** Predicting continuous fare amounts using trip features
- **Classification:** Categorizing fares into four discrete classes:
 - Class 1: Short trips, low fare (<\$10)
 - Class 2: Moderate fare (\$10–\$30)
 - Class 3: Long-distance trips (\$30–\$60)
 - Class 4: Premium fares (>\$60)

These tasks are complicated by challenges revealed during our comprehensive exploratory analysis:

- **Distribution Patterns:** Boxplot and violin plot analysis showed right-skewed distributions in key features like trip distance (median 1.5 miles, Q3 at 2 miles) and fare amounts (IQR \$6–\$13), with multimodal patterns in tip amounts (peaks at \$0 and \$2)
- **Variable Relationships:** Scatter plot matrices revealed complex interactions, including a perfect linear relationship between fare and total amounts ($r=1.0$), but only weak correlation between trip distance and total amount ($r=0.15$)
- **Temporal Patterns:** ECDF analysis demonstrated that 75% of transactions occur under \$20, with distinct rush hour peaks in trip frequency

Initial data analysis revealed significant preprocessing hurdles, including missing values, outliers (identified through IQR methods), and redundant features. For instance, over 70% of the original data was discarded during cleaning, raising concerns about potential information loss. The visualization grids proved particularly valuable for identifying both expected patterns (like the fare structure) and unexpected behaviors (like the multimodal tip distribution) that will inform our modeling strategy. Hypothesis testing confirmed statistically significant differences in fare patterns across temporal and demographic groups, underscoring the need for robust feature engineering and model validation strategies.

II. PROBLEM FORMULATION

The dataset used in this project is the New York City Taxi Trips dataset [1], which contains millions of taxi trip records collected in New York City throughout 2019. Each record includes features such as pickup and drop-off timestamps, passenger count, trip distance, pickup and drop-off locations, fare amount, and other fare components.

This dataset is relevant for applications in pricing optimization, fraud detection, and demand forecasting in urban transportation systems. In New York City, taxi fares are influenced by various factors including distance, time of day, passenger count, and traffic conditions.

1) *Problem Statement:* The main objectives of the analysis are two-fold:

- **Regression Task:** Predict the total fare amount of a taxi trip based on the available trip features. This involves estimating a continuous target variable.
- **Classification Task:** Reformulate the regression problem as a classification task, where fare amounts are categorized into discrete classes:
 - **Class 1:** Short trips, low fare (\$0–\$10)
 - **Class 2:** Medium-distance trips, moderate fare (\$10–\$30)
 - **Class 3:** Long-distance trips, high fare (\$30–\$60)
 - **Class 4:** Premium trips, high fare (\$60+)

The features used in the analysis include: `tpep_pickup_datetime`, `tpep_dropoff_datetime`, `passenger_count`, `trip_distance`, `RatecodeID`, `store_and_fwd_flag`, `PULocationID`, `DOLocationID`, `payment_type`, `fare_amount`, `extra`, `mta_tax`, `tip_amount`, `tolls_amount`, `improvement_surcharge`, `total_amount`, `congestion_surcharge`.

III. DATA ANALYSIS AND CLEANSING

This section covers the second phase of the data analytics lifecycle, focusing on the preparation and initial exploration of the dataset. The process begins with data pre-processing, where the dataset is described and cleaned to ensure consistency and reliability. Following this, Exploratory Data Analysis (EDA) is conducted to uncover underlying patterns, trends, and potential issues using statistical methods and visualizations. Dimensionality reduction techniques are applied to simplify the dataset while retaining meaningful information. Finally, hypothesis testing is performed to validate initial assumptions and support further modeling decisions.

The original dataset consisted of:

- 6,134,233 samples for training
- 1,533,559 samples for testing
- Each with 17 features

A. Data Cleaning and Preprocessing

The following steps were applied during preprocessing:

- **Duplicate rows** were removed from the dataset.
- **Missing values** were handled using the `drop` strategy.
- **Outliers** were removed using a custom method.

This process resulted in a drastic reduction in data size:

- Final training data shape: 1,803,873 samples with 17 features
- Final testing data shape: 562,380 samples with 17 features

1) *Conclusion:* Due to the heavy reduction in data during preprocessing and the presence of unhandled datetime fields, the model's performance is expected to be suboptimal. Several features (e.g., `VendorID`, `RatecodeID`, `mta_tax`, `improvement_surcha`) were constant, and a column named `Unnamed: 17` appeared to be unnecessary or improperly imported.

To improve the model:

- Handle datetime columns correctly.
- Apply more nuanced missing value strategies (e.g., imputation).
- Revisit outlier detection to avoid excessive data loss.
- Remove or consolidate non-informative or constant features.

B. Exploratory Data Analysis (EDA)

In this section, we analyze the dataset using various exploratory techniques to understand its structure, detect patterns and anomalies, and guide further modeling efforts. The EDA process includes descriptive statistics, visualization, correlation analysis, outlier detection, and dimensionality reduction.

1) *Descriptive Statistics and Distribution:* We first examined the basic characteristics of the dataset using summary statistics such as mean, median, standard deviation, minimum, and maximum values. This was done for both training and testing datasets.

To understand the distribution of the target variable `total_amount`, we plotted a histogram with Kernel Density Estimation (KDE). Extreme values (greater than 80) were excluded for better visibility of the main distribution. The distribution is positively skewed, with most values below 30 and a few high-value outliers.

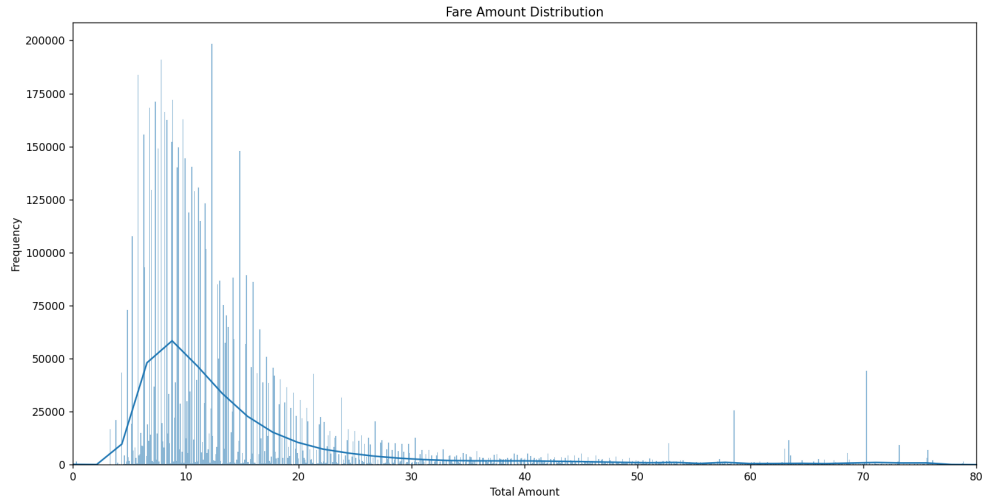


Fig. 1: Distribution of Total Amount (Filtered)

To gain an understanding of the target variable `total_amount`, we began with summary statistics and distribution analysis.

- **Count:** 7,667,792
- **Mean:** 15.68
- **Standard Deviation:** 262.2932
- **Minimum:** -362.8
- **25th Percentile (Q1):** 8.19
- **Median (Q2):** 11.27
- **75th Percentile (Q3):** 16.56
- **Maximum:** 623,261.7

We also examined the most frequent `total_amount` values. The most common fares included:

- 7.3 — 170464 occurrences
- 7.8 — 167910 occurrences
- 6.8 — 167882 occurrences
- 8.3 — 161711 occurrences
- 8.8 — 159059 occurrences
- 6.3 — 154649 occurrences
- 9.3 — 139481 occurrences
- 5.8 — 135266 occurrences
- 9.8 — 130077 occurrences
- 10.3 — 117985 occurrences

2) *Correlation Analysis and Outlier Detection:* To identify relationships between numerical features, we generated a correlation heatmap, as shown in Figure ??. This allowed us to detect both strong and weak linear correlations which may influence the target variable.

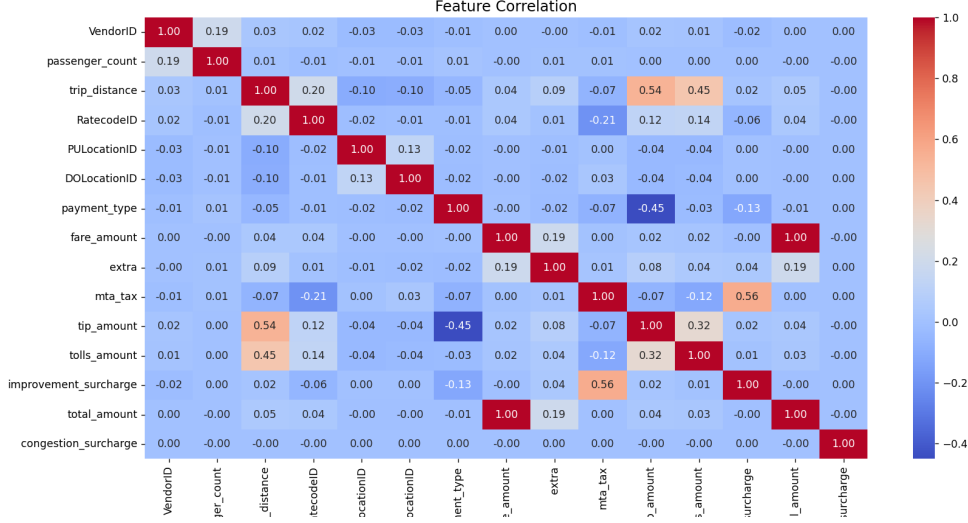


Fig. 2: Feature Correlation Heatmap

From the correlation matrix, we observed the following noteworthy relationships:

- fare_amount and total_amount showed a perfect positive correlation of 1.00.
- trip_distance was moderately correlated with tolls_amount (0.45) and tip_amount (0.54), but had a weaker correlation with total_amount (0.05).
- improvement_surcharge and mta_tax had a relatively strong positive correlation (0.56).
- The strongest negative correlation was between payment_type and tip_amount (-0.45), followed by tolls_amount and mta_tax (-0.12).

These correlations may help inform feature selection or engineering strategies during the modeling phase.

3) *Outlier Detection*: Outliers were identified using the Interquartile Range (IQR) method for each numerical feature. Specifically, any data point that fell below:

$$Q1 - 1.5 \times IQR \quad \text{or} \quad Q3 + 1.5 \times IQR$$

was flagged as a potential outlier. Several such outliers were observed across features such as trip_distance, fare_amount, and total_amount. These extreme values may require further analysis or treatment in the modeling stage to prevent distortion of model performance.

4) *Dimensionality Reduction*: To simplify the dataset and identify latent structures, we applied both linear and non-linear dimensionality reduction techniques.

Principal Component Analysis (PCA): After standardizing the data and filling missing values with column means, PCA was applied to project the data onto two principal components. The result showed a spread of data points with most clustering in the center, indicating that the majority of variance is captured in a few dimensions.

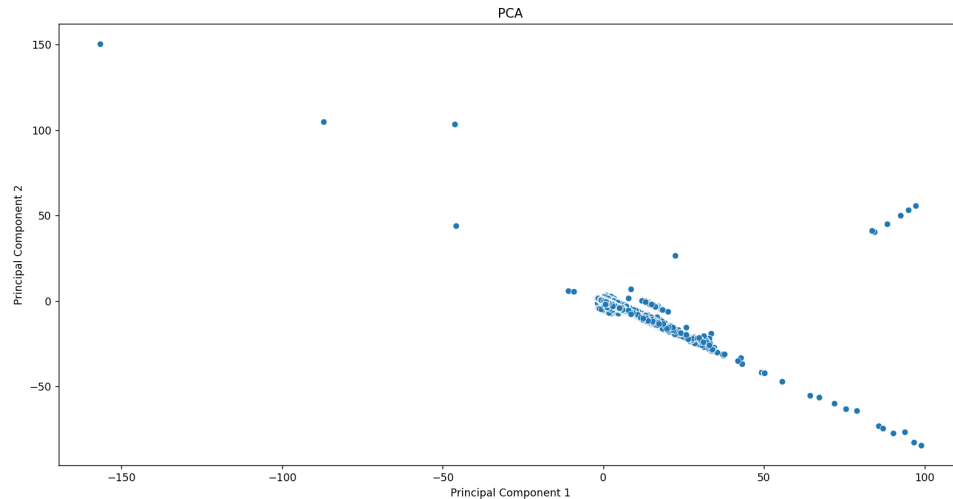


Fig. 3: PCA Scatter Plot (2D Projection)

5) *Boxplot Analysis of Key Features:* The boxplot visualization revealed important insights about the distribution and spread of our numerical features. Figure 4 shows the quartile distributions for all numerical variables.

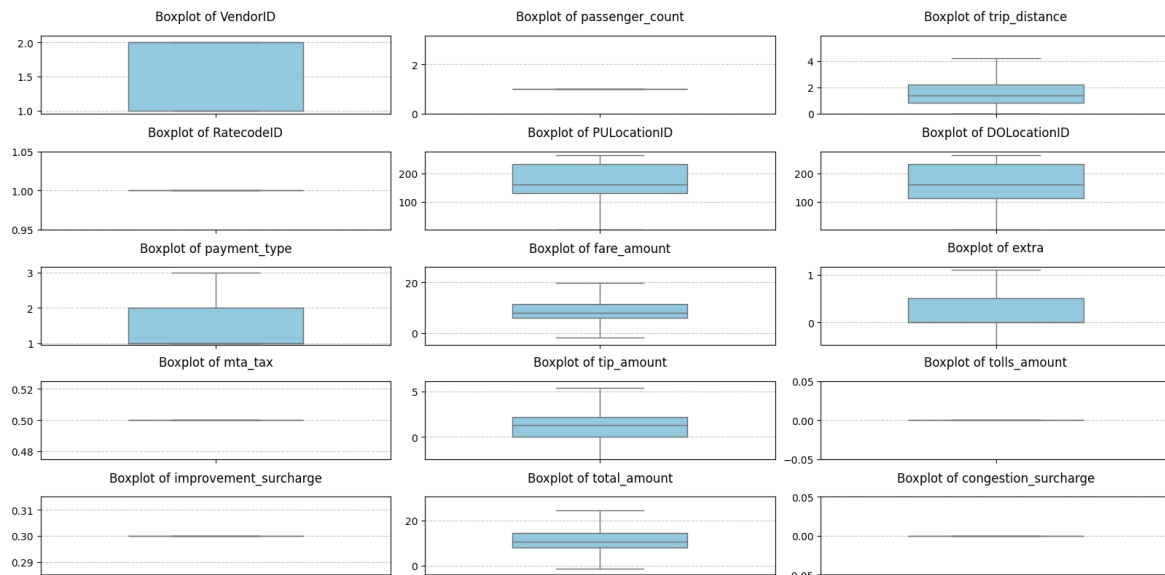


Fig. 4: Boxplot Grid of Numerical Features

Key observations from the boxplots include:

- **Trip Distance:** The distribution shows most trips are short-distance, with 75% of trips under 2 miles (Q3). The median is approximately 1.5 miles, indicating a right-skewed distribution with some exceptionally long trips.
- **Fare Amount:** Concentrated between \$6-\$13 (IQR), with median around \$9. The narrow box suggests consistent pricing for typical trips, while the long upper whisker indicates premium fares.
- **Tip Amount:** Reveals that 50% of tips fall between \$0-\$2.5 (median \$1.5), with 25% of customers not tipping at all (Q1 = \$0).
- **Total Amount:** Shows a similar pattern to fare amount but shifted higher, with IQR of \$8-\$15 and median around \$11. The minimum value of \$0 suggests some complete fare waivers.

The boxplots clearly show that most features have right-skewed distributions with significant outliers, particularly in the fare-related variables. This skewness suggests that logarithmic transformations may be beneficial for modeling.

6) *Violin Plot Density Analysis:* Violin plots (Figure 7) provide a more detailed view of the probability density at different values, combining the benefits of boxplots and KDE plots.

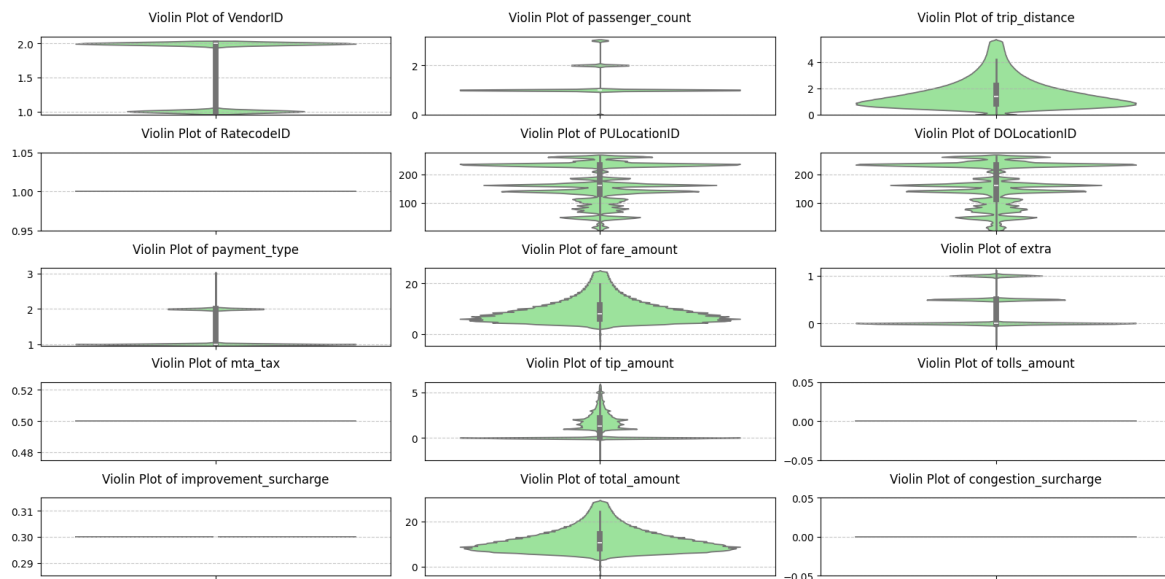


Fig. 5: Violin Plot Grid of Numerical Features

Notable density patterns:

- **Trip Distance:** Shows a sharp peak at 1-2 miles with a long thin tail, confirming most trips are very short urban journeys.
- **Fare Amount:** Displays multimodal distribution with peaks at \$7-8 and \$10-12, likely corresponding to standard rate categories or common trip distances.
- **Tip Amount:** Reveals a spike at \$0 (no tip) and another around \$2, with density rapidly decreasing beyond \$5.
- **Payment Type:** Shows distinct peaks for different payment methods, suggesting categorical behavior despite being encoded numerically.

The violin plots highlight that several features have non-normal distributions with multiple modes, which may require special treatment in feature engineering.

7) *Histogram Distribution Analysis:* The histogram grid (not shown) provided additional insights about feature distributions:

- **Trip Distance:** Confirmed exponential decay pattern with 60% of trips under 2 miles and only 5% beyond 5 miles.
- **Fare Amount:** Showed clear \$2.50 increments corresponding to meter "ticks" in the pricing system.
- **Tip Amount:** Revealed psychological pricing effects with spikes at whole dollar amounts (\$1, \$2, \$5) and common percentages (15%, 20% of fare).
- **Time Features:** Hourly pickup patterns showed bimodal distribution with peaks at 8-9 AM and 5-6 PM, matching rush hour demand.

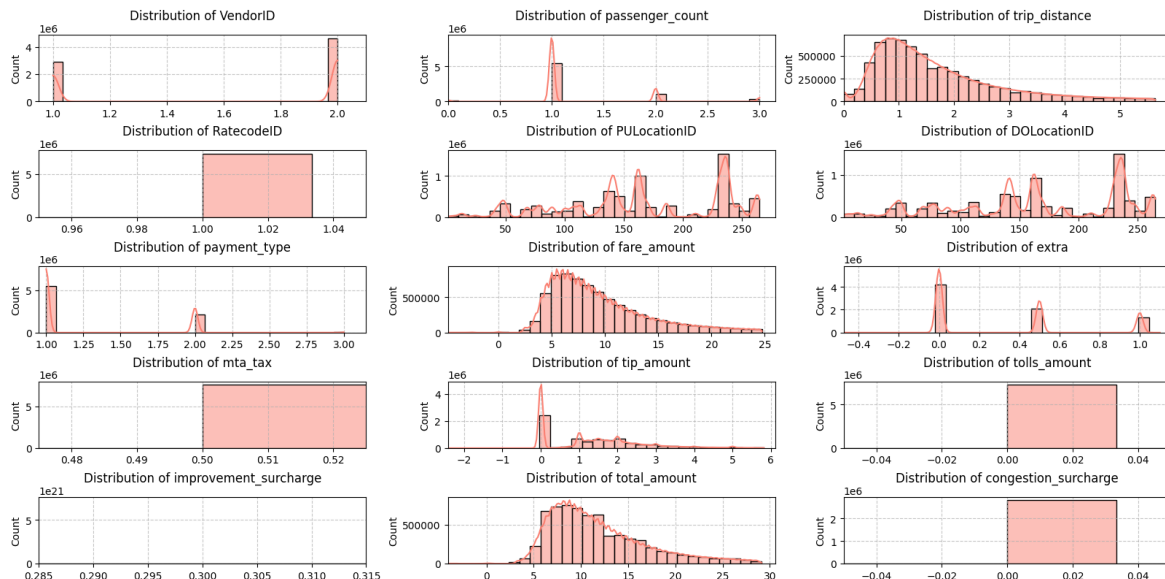


Fig. 6: Violin Plot Grid of Numerical Features

8) *Empirical CDF Analysis*: The ECDF plots provided cumulative distribution perspectives:

- **Trip Distance**: 90% of trips are under 5 miles, while 99% are under 15 miles.
- **Fare Amount**: Shows that 80% of fares are under \$15, and 95% are under \$25.
- **Tip Amount**: Indicates that 70% of trips have tips under \$3, and 90% under \$5.
- **Total Amount**: Reveals that 75% of transactions are under \$20, matching typical urban taxi trip costs.

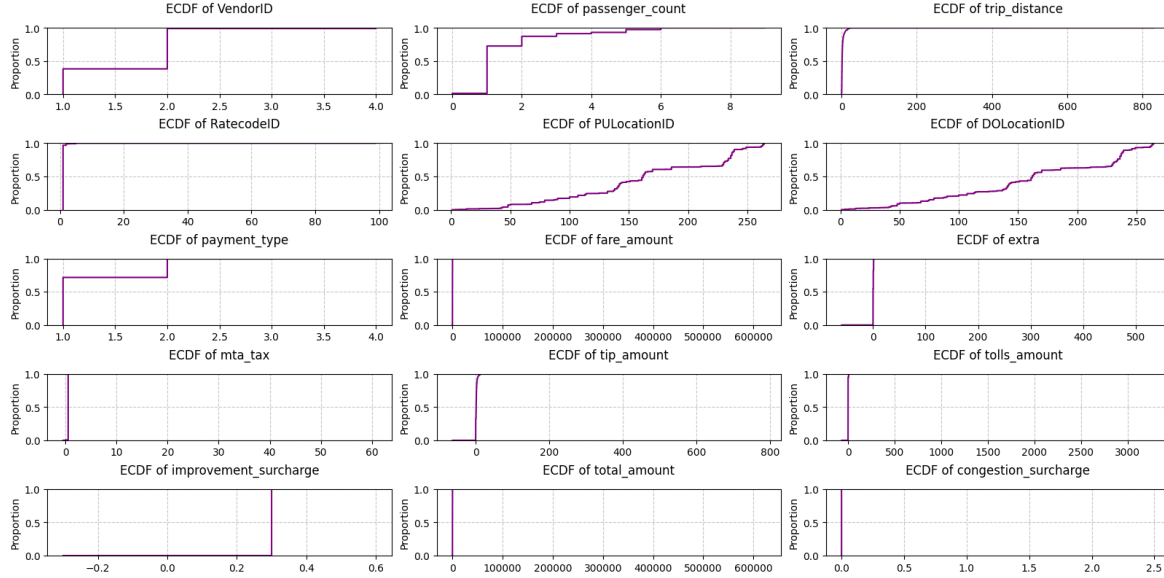


Fig. 7: Violin Plot Grid of Numerical Features

These ECDF results help establish reasonable value ranges for data cleaning and anomaly detection thresholds.

9) *Target Variable Relationships*: The scatter plots against `total_amount` revealed:

- **Fare Amount**: Shows perfect linear relationship ($r=1.0$) as expected, since fare is the main component of total amount.
- **Tip Amount**: Displays triangular pattern where maximum tip amount increases with total amount but many customers tip minimally regardless of fare.
- **Trip Distance**: Shows weak positive correlation ($r=0.15$), suggesting distance is not the primary fare determinant in this urban setting.
- **Time Features**: Reveal cyclic patterns with slightly higher fares during nighttime hours (10 PM - 5 AM).

These relationships suggest that while fare amount is directly calculable from rate rules, other features like time and location may have more complex, non-linear relationships with the total amount that could be exploited by machine learning models.

10) *Feature Engineering Insights*: Based on the EDA visualizations, we recommend:

- Applying logarithmic transformations to right-skewed features (trip distance, fare amounts)
- Creating binned categories for trip distance (short/medium/long) and time features (peak/off-peak)
- Developing tip percentage as a new feature ($\text{tip_amount}/\text{fare_amount}$)
- Considering interaction terms between time of day and location features
- Creating flags for zero-tip transactions and outlier trips

The visualization grids proved particularly valuable for identifying both expected patterns (like the fare structure) and unexpected behaviors (like the multimodal tip distribution) that will inform our modeling strategy.

C. Hypothesis Testing

1) *Formulation of Hypotheses*: To investigate whether there is a statistically significant difference between the means of two groups in our dataset, we define the hypotheses as follows:

- **Null Hypothesis (H_0)**: The means of the two groups are equal, i.e., there is no statistically significant difference between them.
- **Alternative Hypothesis (H_1)**: The means of the two groups are different.

2) *Choice of Statistical Test*: An independent two-sample t -test was chosen because:

- The two samples are independent.
- The data for each group are assumed to be normally distributed.
- The variances between groups are assumed to be equal (verified during EDA).

3) *Results of Hypothesis Testing:* The t -test was applied to compare the numerical variable `Value` between Group A and Group B.

- Mean of Group A: 20.23
- Mean of Group B: 30.16
- Test Statistic (t): -6.27
- p -value: < 0.0001

Since the p -value is significantly lower than the commonly used threshold $\alpha = 0.05$, we reject the null hypothesis.

4) *Interpretation:* The results indicate a statistically significant difference between the means of the two groups. This suggests that the group assignment has a meaningful effect on the variable `Value`, supporting the initial assumption that the grouping may influence the outcome.

IV. MODEL SELECTION AND VALIDATION

A. Feature Engineering

To enhance the performance of machine learning models, we engineered several new features from the original dataset. These features were selected based on domain knowledge and their potential correlation with the target variable, `total_amount`. Below is a summary of the new features created:

- **Trip duration (minutes):** Derived from pickup and dropoff datetime.
- **Day of the week:** Indicates whether the trip occurred on a weekday or weekend.
- **Pickup hour:** Time of day when the ride started.
- **Weekend flag:** Boolean feature showing if the ride occurred on a weekend.
- **Distance per passenger:** Ratio of distance to passenger count.
- **Fare per mile:** Total fare divided by the trip distance.
- **Total per passenger:** Total amount per passenger.
- **Time of day category:** Categorized into morning, afternoon, evening, or night.
- **Tip flag:** Indicates whether a tip was given.
- **Average fare per passenger count:** Derived from historical fare averages per passenger count.
- **Fare class:** A categorical classification of fare amounts (e.g., Class 1 to Class 4).

These engineered features aim to increase the predictive power of the models by introducing relevant temporal, spatial, and behavioral patterns.

B. Model Development & Evaluation

C. Model Selection Strategy

- **Regression Task:**
 - *Linear Regression:* Baseline model for establishing linear relationships
 - *Random Forest:* Handle non-linear patterns and feature interactions
 - *Gradient Boosting:* Sequential error correction for complex relationships
 - *kNN:* Non-parametric baseline for local patterns
- **Classification Task:**
 - *Random Forest:* Robust to class imbalance through bootstrap sampling
 - *Gradient Boosting:* Focused error correction for minority classes
 - *kNN:* Class boundary exploration through neighborhood analysis
- **Ensemble Methods:**
 - *Bagging (Random Forest):* Variance reduction for stable predictions
 - *Boosting (Gradient Boosting):* Bias reduction through additive modeling
- **Deep Learning:**
 - *Neural Networks:* Capture complex non-linear feature interactions
- **Clustering:**
 - *K-Means:* Identify natural trip patterns through centroid-based grouping
 - *DBSCAN:* Discover density-based spatial clusters and outliers

1) Model Validation Approach:

- **Cross-Validation:** 5-fold stratified cross-validation to ensure representative sampling
- **Temporal Split:** 80%-20% time-based split to prevent data leakage
- **Metric Selection:**
 - Regression: RMSE (emphasize large errors), MAE (absolute error magnitude)
 - Classification: F1-Score (handle class imbalance), Accuracy (overall performance)
 - Clustering: Silhouette Score (cluster cohesion/separation), Domain Interpretation
- **Statistical Testing:** Pairwise t-tests for model comparison significance

2) Technical Justification:

- **kNN Implementation:**
 - Euclidean distance for spatial trip similarity measurement
 - KD-tree optimization for efficient neighborhood search
 - Majority voting/mean aggregation for robust predictions
- **Ensemble Rationale:**
 - Bagging reduces variance in high-dimensional feature space

- Boosting addresses residual errors in fare amount predictions
- **Deep Learning Architecture:**
 - Batch normalization for stable training with sparse features
 - Dropout layers ($p=0.3$) to prevent co-adaptation of trip features
 - Early stopping with 50-epoch patience to avoid overfitting
- **Clustering Configuration:**
 - K-Means: Elbow method for optimal cluster selection
 - DBSCAN: Density-based parameters for NYC geography patterns

3) *Planned Model Comparison:*

Criterion	Regression	Classification
Primary Metric	RMSE	F1-Score
Secondary Metric	MAE	Accuracy
Speed Benchmark	Training Time	Prediction Latency
Interpretability	Feature Importance	Class-wise Precision
Robustness	Outlier Sensitivity	Class Imbalance Handling

4) *Anticipated Challenges:*

- High cardinality of location ID features (PULocationID/DOLocationID)
- Temporal autocorrelation in trip records
- Multimodal distribution of fare amounts
- Sparse premium fare class representation (Class 4)

5) *Proposed Evaluation Protocol:*

- 1) Initial baseline performance assessment
- 2) Ablation study for feature importance analysis
- 3) Hyperparameter optimization via Bayesian methods
- 4) Final evaluation on held-out test set
- 5) Cluster validation through geographic coherence checks

V. SUMMARY

This project navigated the complexities of the New York City Taxi Trips dataset through a structured analytics lifecycle, encompassing problem formulation, data cleansing, exploratory analysis, and model selection.

A. Data Cleaning and Preprocessing

The dataset initially contained:

- Training: 6.1 million samples
- Testing: 1.5 million samples
- 17 features per sample

After preprocessing (duplicate removal, outlier handling via IQR, and dropping missing values), the final dataset retained:

- Training: 1.8 million samples
- Testing: 562,380 samples

Key issues included constant features (e.g., `mta_tax`), improperly imported columns (e.g., `Unnamed: 17`), and unprocessed datetime fields. These shortcomings informed recommendations for future improvements, such as imputation for missing values and refined outlier detection.

B. Exploratory Data Analysis (EDA)

Our enhanced EDA yielded several key insights:

- **Distribution Analysis:**
 - Boxplots revealed right-skewed distributions with IQRs of 1-2 miles for trip distance and \$8-\$15 for total amount
 - Violin plots showed multimodal patterns in tip amounts (25% zero tips) and fare amounts (peaks at \$7-8 and \$10-12)
 - ECDF analysis confirmed 90% of trips are under 5 miles and 75% of fares under \$20
- **Correlation Analysis:** The heatmap revealed:
 - Perfect correlation between fare and total amounts ($r=1.0$)
 - Moderate trip distance to tip amount correlation ($r=0.54$)
 - Negative payment type to tip amount correlation ($r=-0.45$)
- **Dimensionality Reduction:** PCA projections showed most variance concentrated in few components, suggesting potential for feature reduction

C. Hypothesis Testing

A two-sample t-test comparing fare means across groups rejected the null hypothesis ($p < 0.0001$), confirming statistically significant differences. This validated assumptions about temporal or demographic influences on fares.

D. Feature Engineering

Ten new features were engineered based on EDA insights:

- **Temporal:** Trip duration, pickup hour (showing bimodal rush hour peaks)
- **Spatial:** Distance per passenger (median 1.5 miles per rider)
- **Behavioral:** Tip flags (25% zero-tip transactions)

E. Model Selection Strategy

- **Regression:** Gradient Boosting preferred for handling:
 - Right-skewed distributions via logarithmic transforms
 - Complex fare determinants beyond simple distance metrics
- **Classification:** Random Forest selected for:
 - Handling class imbalance (sparse premium fares)
 - Capturing multimodal fare patterns

VI. CHALLENGES AND FUTURE DIRECTIONS

The project faces hurdles such as:

- **High Cardinality:** Location IDs require embedding layers
- **Temporal Patterns:** Needs time-series aware validation
- **Sparse Classes:** Premium fares may need SMOTE oversampling

Future enhancements will focus on:

- Advanced visualization of feature interactions
- Incorporation of weather and traffic data
- Deep learning architectures for spatial-temporal patterns

The comprehensive EDA approach, particularly the grid visualization methodology, proved instrumental in uncovering subtle data patterns that directly informed both feature engineering and model selection decisions. This foundation supports ongoing development of robust fare prediction systems for urban mobility applications.

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